MILITARY SPECIFICATION

SHOP EQUIPMENT ELECTRONIC REPAIR,

SEMITRAILER MOUNTED

This specification is approved for use by the Armament Research, Development and Engeineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mobile, semitrailer-mounted electronic repair shop.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise speci fied the issue of them documents are those l isted in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATION

FEDERAL

FF-P-101	-	Padlock
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
W-F-800	-	Fuel Oil, Diesel
PPP-B-s66	-	Box, Folding, Paperboard
PPP-B-601	-	Box, Wood, Cleated-Plywood

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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PPP-B-636	Box , Shipping, F i berboard
PPP-B-1055	Barrier Material, Waterproofed, Flexible
PPP-H-1581	Hardware (Fasteners and Related Items): Packaging
PPP-P-40	Packaging and .Packing of Hand Tools
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture and Fungus-Resistant (For Treatment of Communication, Electronic and Associated Equipment)
MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-T-740	Truck, 5 Ton. 6 x 6. Military Design M39 Series
MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-L-2105	Lubricating Oil, Gear, Multi-purpose
MIL-C-5541	Chemical Conversion Coatings for Aluminum and Aluminum Alloys
MIL-M-7866	Molybdenum disulfate, Technical Lubrication
MIL-E-10062	Engine, Preparation for Shipment and Storage of
MIL-G-10924	Grea se, Automotive and Artillery
MIL-C-12000	Cable, Cord and Wire, electric; Packaging of
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-M-18058	Machinery, Metal and Wood-Working, Support Equipment and Associated parts; Preparation for Delivery of
MIL-D-23377	Primer- Coating, Epoxy-Polyamide Chemical and Solvent Resistant
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-G-28554	Generator Set, Mobile Electric Power and Supplemental Equipment ; Packaging of
MIL-W-46567	Welding and soldering equipment Supplies, and A ccessories: .Packaging of
MIL-L-46157	Lubricating oil, Internal Combustion Engine, Arctic

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MIL-B-46176 -	Brake Fluid, Silicone, Automotive All Weather, operational and Preservative
MIL-G-52732/''ll -	Generator Set, Gasoline Engine-Driven, 10 KW, 60 Hertz and 28 Volt Direct Current, Type I (Tactical) Class 2 (utility) General Specification for
MIL-C-53072	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-V-62049	Vehicles, Wheeled: Preparation for Shipment, and Storage of
MIL-S-62049	Semitrailer, Van: For Installation of Special equipment, Military Vehicles

STANDARDS

MILITARY

MIL-STD-129	%rking for Shipment. and Storage
MIL-STD-130	Identification Marking of US Military property
M IL STD- 1 7 1	Finishing of Metal and Wood Surfaces
MIL-STD-454	Standard General Requirements for Electronic Equipment, Requirement 4
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and waterproofing; With Appropriate Test Methods
MIL-STD- 11 90	Minimum guidelines for Level C Preservation, Packing and Marking
MIL-STD-1471	Noise Level for Army Materiel

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center. (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF THE ARMY

DRAWINGS

[illegible]	- List of Components, Electronic Shop Set
TA13017E8200	- Shop Equipment, Electronic Repair, Semitrailer Mounted

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(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the document cited in the solicitation (see 6.2.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Salt Spray (FOG) Testing
ASTM D3359 - Meausring Adhesion by Tape Test

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection in accordance with 4.3. the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification,

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3.2 Description. The repair shop shall be in accordance with Top Assembly Drawing TA13217E8200 and as specified herein. The shop shall include all tools and equipment specified in the Set Components List as compiled from drawing A7551983 (see 6.3).

3.2.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the shop equipment. Any data, such as shop drawings, layouts, flow sheets, processing procedures, etc., prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.3 Initial production. When specified (see 5.2.1), the contractor shall furnish to the Government one or more initial production electrical repair shops for inspection.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free from defects which would adversely affects the performance or maintainability of the individual components or overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.5 Shop dimensions.

3.5.1 Dimensions. Dimensions of the shop shall be as shown on the applicable drawings (see 3.2).

3.6 Body. The contractor shall Fabricate and install the equipment on the chassis in accordance with the applicable drawings (see 3.2). The contractor shall furnish all items of equipment including the tools and equipment as specified in 3.2.

3.6.1 Padlock. The contractor shall furnish and install on the rear door and ladder a padlock with hardened shackle and chain in accordance with FF-P-101, type EPA, size 1 1/8 inch. The free end of the chain shall be riveted to the door.

3.7 Electrical system. Electrical system shall be in accordance with the applicable drawings and comply with the NFPA 70.

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3.8 Tools and equipment. The tools and equipment, in their as-used condition, shall be stored and anchored to prevent movement during transit. Instruments which do not have transit cases and which are sensitive to damage by vibration shall be protected in their storage compartments by polyurethane closed-cell foam sheet. Tiedown devices shall be designed for a general tool or equipment configuration to allow interchange of like items from different supply sources. Wherever possible, items shall be grouped together by function.

3.9 Safety All exposed parts, assemblies, or accessories which are subject to high operating temperatures, or which are energized electrically, shall be fully insulated, enclosed, or guarded. All moving parts which are hazardous to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions of the shop .

3.10 Performance.

3.10.1 installed equipment and power tools operation. Using both external and shop generated power, operate all installed equipment and tools.

3.10.2 Roadability. When equipped with the tools and equipment specified on drawing A7551983 and loaded in accordance with 3.8, the shop shall withstand towing by a Government loaned M818 truck tractor (see 3.20) on unimproved roads, cross country trails, a chatter course and a diagonal ditch course. During or following towing, there shall be no damage to, or displacement (see 6.8.2) of any component, accessory, part or tool installed in or on the shop and no evidence of defective welded, bolted or riveted connections on the shop.

3.10.3 Rail impact, The shop equipment with all tools and equipment stored in the proper place and with all panels and doors closed, shall withstand a rail impact test in accordance with MIL-STD-810, without damage to the shop or the contents, when impacted at velocities of 4, 6, and 8 miles per hour (mph).

3.11 Environmental conditions.

3.11.1 Heating system. operate the heaters (see 3. +19 continuously for 15 minutes to check its performance.

3.11.2 Air conditioner. Operate the air- conditioner (see 3.19) continuously for 15 minutes to check its performance.

3.11.3 Temperature (storage and operating).

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3.11.3.1 Storage. The shop with all equipment, shall be capable of withstanding storage temperatures down to -50, plus/minus 5, degrees Fahrenheit (F) without encountering any adverse effects or damage. The complete shop shall be capable of operation when returned and stabilized to an ambient temperature within the operating range.

3. 11.3.2 Operational temperature(s). The shop shall perform as specified below in any ambient temperature between -25, plus/minus 2, degrees F and +120, plus/minus 5, degrees F:

a. Following a 24 hour soak period at -25, plus/minus 2, degrees F and while at that temperature, operate the installed heater, all portable electric power tools, and electronic equ ipment from the shop installed p ower sour ce. Slave starting of the government furnished generator may be attempted if starting difficulties are encountered (see 6.8.8). When returned to ambient temperature, the complete shop shall be operated.

b. Following a 24 hour soak period at +120, plus/minus 5, degrees F and while at that temperature, operate all shop equipment except the installed heater from the shop installed power source.

3.11.4 Sound Level. The sound level of the shop Shall be in accordance with MIL-STD-1474 Category D for steady-state noise when operating shop equipment. Hearing protectors need not be supplied with the shop equipment. Hazard signs, if required, shall state: "Caution: Hearing .Protection Required" (see 6.12).

3.12 Reliability. The shop equipment shall demonstrate a mission reliability of at least 0.90 when measured with mean time between failures (MTBF).

3.13 Availability. The shop equipment availability shall be not less than 95 percent.

3.14 Maintainability. The shop equipment shall require a minimum of operational and in-storage maintenance. Replacement components shall be used to the maximum extent possible. The shop equipment shall require a minimum of man-hours and skill to accomplish maintenance. Ease of maintenance and restricted use of components requiring special skills or tools for maintenance shall be stressed. The maintenance ratio shall be no greater than 0.12 (see 6.8.1).

3.15 Fuels, lubricants, and coolants. The shop(s) shall be assembled and tested with the products of the type and grade specified in table I for the applicable ambient temperatures. The shops shall operate as specified herein when furnished with these products (see 6.5).

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TABLE I. Service Products

tern	Application	Specification	Type & Grade
1.	Engine:		
	(a) Heavy Duty	MIL-L-2104	10, 30
	(b) Arctic	MIL-L-46167	One Grade
2.	Fue 1:		
	Diesel Fuel	VV-F-800	
3.	Grease:		
	(a) Automotive	MIL-G-19924	One Grade
	(b) High Temperature	MIL-C-23827	One Crade
4.	Hydraulic-system Oil:		
	(a) Oil	MIL-L-2104	10
	(b) Oil (lOW temperature)	MIL-L-45167	One Grade
5	Lubricating Oils:		
	Universal	MIL-L-2105	80, 90, 140
6.	Brake Fluid	MIL-B-46176	one Grade

3.15.1 Oil lubrication. Unless otherwise specified herein, all components all be filled to the operating level with military oils designated for use in e temperature range at the shipping destination. A tag shall be attached in conspicuous place to indicate the temperature range and grade of lubricant ed.

3.16 Fungus resistance. Electrical and Electronic components and ectrical insulation not otherwise specified herein shall be fungus inert, in cordance with requirement 4 of MIL-STD-454. Components Or electrical rcuitry that are not weather resistant to moisture and fungus growth shall be eated in accordance with MIL-T-152 and MIL-V-173.

3.17 Identification marking and instruction plates.

3.17.1 Electromagnetic Interference (EMI). A permanent warning notice, dicating that EMI could result if electric power tools/equipment are operated close vicinity to electronic communication or surveillance equipment, shall included and displayed in a conspicuous place(s) on the shop van exterior trance(s) and on the master power control panel.

3.17.2 Identification marking, .The shop shall be identified in accordance th MIL-STD-130.

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3.17.3 Instruction plates. Instruction plates shall conform to MIL-P-514, Type III, Composition C, or Type I, Style 1, Composition A, Class 2.

3.18 Treatment and painting. The portions of the shop normally painted shall be cleaned, treated, and painted in accordance with Finish 5.1.1 for ferrous surfaces-and Finish 7.3.1 for non-ferrous surfaces plus Finish 20.24 for exterior areas and Finish 24.17 for interior areas as specified in MIL-STD-171 and as specified on the detail drawings. Application shall be in accordance with MIL-C-53072.

3.19 Government furnished property (GFP). Unless otherwise specified (see 6.2.1), the following property in the quantities indicated shall be furnished by the Government (see 6.7).

Item No.	Description	Identification	Quantity for each shop
1	Semitrailer, Van, Shop, Folding Side, 6-Ton, 4-Wheel, with air conditioner and heaters.	M313 MIL-S-62049 EPL 8736316	1

3.20 Government loaned property (GLP). Unless otherwise specified (see 6.2.1), the following property in the quantities indicated shall be loaned by the Government (see 6.6);

Item No.	Description	Identification	Quantity
1	Truck-tractor, M818 NSN 2320-00-050-8984	MIL-T-740	1
2	Generator Set, Gas Engine 10KW, AC, 120/208V., Three-Phase NSN 6115-00-889-1447	MIL-G-52732 MEP018A	1

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3.21 Workmanship. Standards of workmanship shall be such that the shop shall have the safety, stability, and operating characteristics specified herein and shall be free of defects that could adversely affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor my use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor shall be responsible for ensuring that components and materials used are manufactured, tested, and inspected in accordance with referenced specifications and standards.

4.1.3 Preparation for examination and tests. Prior to examination and test, the shop shall be lubricated with oils and greases specified in 3.15, table I. Oils shall be those designated for use in the ambient temperature at which the examinations and tests are being performed.

4.1.4 Examination or test of GFP or GLF items. Failure of GFP (see 3.19) or Glp (see 3.20 during examinations or tests does not constitute failure of the examination or test. This includes failure of GFP or GLP engines to start during the tests. Slave starting may be used. Failure of GFP and GLP items are to be reported through normal channels.

4.2 Classification of inspection. The inspection requirements specified herein are:

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a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Packaging inspection (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined for the defects marked "X" in column 3 of table II. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.3.3.

TABLE II. Examination schedule

Defect	Requirement Paragraph	First Article	Initial Production	Quality Conformance
1	2	3	4	5
Assembly incomplete or incorrect	3.2	X	X	X
Material not as specified	3.4	X	-	X
Dimensions not as specified	3.5	X	-	-
Body and equipment not as specified	3.6 thru 3.7, 3.19	X	X	X
Parts, components, tools or equipment missing or as specified	3.8	X	X	X
Stowage or tiedowns not as specified	3.8	X	X	-

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TABLE 1 I . Examination schedule (Continued)

Safety devices and guards missing or not as specified	3.9	x	x	x
Lubrication not as specified.	3.15, 3.15.1	x	x	x
Fungus and moisture resistance not as specified	3.16	x		x
Identification marking, warning and instructions, notices and plates net as specified	3.17	x	x	x
Treatment or painting not as specified	3.18	x	x	x
Workmanship not as specified	3.21	x	x	x

4.3.2 Tests, Each repair shop shall be tested as specified in column 1 of table III. Failure of any test shall be cause for performing the inspection specified (see 4.3.3).

4.3.3. Disassembly inspection. 4 first article shop .'ailing a performance test shall be disassembled in the presence of a Government representative to the extent necessary to determine the cause of the failure. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complyng with such requirements shall be rejected. Reassembly with accepted parts and retesting shall be the responsibility of the contractor.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production shops shall be selected at random by the Government from shops being produced from production tooling and will be examined for the defects marked "X" in column 4 of table II and subjected to the tests marked "X" in column 5 of table III to determine conformance to the requirements of this specification The inspection shall be performed by the Government at a site selected by the Government. Acceptance of an initial production shop shall not exclude the remaining shops from the quality conformance inspection and acceptance provisions specified in Section 4. IN addition to any test

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specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production shop to meet any requirement specified herein during and as a result of the examinatio, and tests specified (see 4.4) shall be cause for rejection of the initial production shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accompl ished by the contractor at no cost to the Government on shops previously produced and accepted under the contract. Any deficiencies found as a result of the initial production inspection shall be considered prima facie evidence that all shops accepted prior to completion of initial production inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is ac ceptable to t he contract ing officer

4.5 Quality conformance inspection.

4.5.1 Examination. Each shop shall be examined for the defects marked X in column 5 of table II. Presence of one or more defects shall be cause for rejection

4.5.2 Tests. Shops shall be subjected to the tests marked X in column 6 of table III. Sampling for tests is to be one of each ten shops for power tool operation (4.6.2.2) and one of each twenty shops for paint finish (4.6.2.10).

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table II.

4.6.2 Tests.

4.6.2.1 Test schedule. Tests skill be in accordance with table III. Sequence of testing shall be at the option of the test facility, except as specified in the paragraphs herein.

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TABLE III. Test schedule.

Test	Test Paragraph	Requirement Paragraph	First Article	Initial Production	Quality Conformance
1	2	3	4	5	6
Power Tools	4.6.2.2	3.2 and 3.10.1	X	X	X
Roadability	4.6.2.3	3.10.2	---	X	---
Rail impact	4.6.2.4	3.10.3	X	---	---
Environmental	4.6.2.6	3.11	X	---	---
Sound level	4.6.2.5	3.11.4	X	---	---
Reliability	4.6.2.7	3.12	---	X	---
Availability	4.6.2.8	3.13	---	X	---
Maintainability	4.6.2.9	3.14	---	X	---
Treatment and Painting	4.6.2.10	3.18	X	---	X

4.6.2.2 Installed equipment and power tools operation. Using both external and shop generated power, operate all installed equipment and power tools for a period of at least 2 minutes on each power source (see 3.10. 1). Inability of installed equipment or power tools to function properly shall constitute failure of this test,

4.6.2.3 Roadability. Load the shop with the tools and equipment specified herein and perform the following tests, per requirements of 3.10.2.

(a) Unimproved roads. Drive the shop 50 miles on unimproved roads at maximum safe speeds. The course shall consist of poorly maintained gravel and dirt roads with rutting, and occasional potholes.

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(b) cross-country trails. Drive the shop 50 miles on cross-country trails at maximum safe speeds. The course shall consist of natjve soil with rock added as required to maintain passability. Surface conditions shall range from smooth to rough with rutting, potholes, and depressed areas.

(c) Chatter course. Drive the shop over a chatter course until at least 2,600 either upward or downward accelerations of the magnitudes specified herein have been applied to a rear wheel center. The course shall be of such construction that the acceleration magnitudes specified below can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than three cos. The maximum acceleration magnitude specified below shall not be exceeded and accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives in one pass over the course at. least 50 consecutive vertical accelration cycles os any magnitude less than the maximum specified herein. Measure the accelerations by a trial test to establish the required vehicle test speed. 'hen disconnect the recording instruments and complete the test at established speed The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the vehicle test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member, usually an axle or spindle, that has a motion identical to that of the wheel center. The magnitude of the acceleration shall be not less than four Gravity (Gs) nor more than seven Gs.

(d) Diagonal ditch course. Drive the shop over a series of ditches until 600 reverse torsional frames stresses have been applied. The ditch profile shall be such so as to allow vertical wheel travel of 18 inches from crest to root. The ditches shall be at 45 degrees to the centerline of the course and spaced a sufficient distance apart to allow only one rear wheel and one front wheel of opposite sides in the ditches at any one time.

Examine the shop at the completion of each phase of the road test. Damage to, or displacement (see 6.8.2) of any component, accessory, part, or tool; evidence of defective welded, bolted, or riveted connections excluding the Government furnished chassis shall constitute failure of these road tests.

4.6.2.4 Rail impact. The shop shall be subjected to a rail impact test in accordance with MIL-STD-810 (see 3.10.3). Operate all power tools, using shop power for at least five minutes, both prior to and after the rail impact test.

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Test is to be conducted with all equipment stored in the proper locations and all panels and doors closed. Evidence of any damage or failure of any comment to perform as required shall be cause for rejection.

4.6.2.5 Sound levels. Measure the sound level at the work station while operating each of the power tools. Sound level in excess of Category D of MIL-STD-1474 shall require the installation of hazard signs as specified in 3.11.4.

4.6.2.6 environmental.

4.6.2.6.1 Heating system, Operate the heaters continuously for not less than 15 minutes to check the operation of the heaters, thermostat, and heat deflectors (see 3.11.1). Any of the following that occurs as a result of faulty material supplied by the supplier or incorrect installation by the. supplier shall constitute failure of this test:

a. Malfunctioning of deflector or fuel supply line.
b. Smoking, sooting, or back draft.
c. Entrance of fuel vapors or exhaust gases into the interior of the shop.

4.6.2.6.2 Air conditioner. Operate the air conditioner continuously for not less than 15 minutes to check the operation of the air conditioner and thermostat (see 3. 11.2).

4.6.2.6.3 Temperature (storage and operation). Maintain the shop at an ambient temperature of l 50, plus/minus 5 , degrees F for a period not less than 72 hours (see 3.11.3.1). At the end of this period raise the temperature to 25, plus/minus 2 degrees F for a period not less than 12 hours. At the end of this time period and with the temperature at -25, plus/minus 2, degrees F. the following operational tests shall be performed (see 3.11.3.2):

a. Inspect the entire unit for any damage which may have resulted from exposure to temperature extreme.
b. Start and operate the heating system. The heater shall operate as required in 4.6.2.6.1.
c. Start and operate individually all portable electric power tools from the shop installed power source for a minimum of 2 minutes.

After the conclusion of the operating test at -25, plus/minus 2, degrees F, stop all equipment, close up the unit, and raise the ambient temperature to +70, plus/minus 5, degrees F. When stabilized at +70, plus/minus 5, degrees F, start and operate individually all instal led equipment and power tools from the shop installed power source for a minimum of 2 minutes.

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After the conclusion of the operating test at +70, plus/minus 5, degrees F, stop all equipment, close up the unit, and raise the ambient temperature to +120, plus/minus 5, degrees F for a period of 24 hours. After the 24 hours, soak at +120, plus/minus 5, degrees F, start and operate individually all installed equipment and power tools from the shop installed power source for a minimum of 2 minutes. Nonconformance to 3.11 shall be cause for rejection of the unit.

4.6.2.7 Reliability. The initial production shop shall be tested for a total of ten missions to determine conformance to the requirements of 3.12. Data shall be gathered to evaluate the system reliability. Reliability is to be evaluated from MTBF and mission operating time (see 6.8.4 and 6.8.5).

$$\text{Reliability} = e^{-(t/u)}$$

where,

t = mission operating time, hour
u = mean time between failures (MTBF), hour
e = 2.718

The shop shall be considered to have demonstrated the required reliability if zero chargeable reliability failures occur (see 6.8.3).

4.6.2.7.1 Mission operation. A mission shall consist of the following:

a. Drive the fully loaded shop set at maximum safe speeds for ten miles on unimproved roads and ten miles on cross-country trails.
b. Using the instructions furnished with the shop, set up the shop for operation.
c. Operate each installed equipment and power tool for a period not less than 5 minutes one time per mission, except as noted. "
d. Prepare the shop for movement to another location, to begin the next mission.

Nonconformance with 3.12 shall constitute failure of this test.

4.6.2.6 Availability. During the performance of the reliability test as specified in 4. 6.2.7, data shall be gathered to evaluate the system availability. Availability is to be evaluated with respect to operating time and corrective maintenance (see 6.8.5 and 5.8.6).

$$\text{Availability} = \frac{(\quad \text{MTBUMA} \quad)}{(\text{MTBUMA} + \text{MTTR})} \quad (100 \text{ percent})$$

where ,

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MTBUMA = Mean time between unscheduled maintenance actions (unscheduled means corrective maintenance actions).

MTTR = Mean time to repair (use only times associated with corrective maintenance) (see 6.8.7).

The system availability requirement shall conform to 3.13. Noncomformance with 3.13 shall constitute failure to meet the availability requirement.

4.6.2.9 Maintainability. Maintenance operations required during first article and initial production tests shall be gathered to evaluate the system maintainability. Maintainability is to be evaluated with respect to total maintenance man-hours and total operating time during testing.

$$\text{Maintenance Ratio} = \frac{\text{TNMMT}}{\text{TOT}}$$

where,

TNMMT = Total number of man-hours of maintenance (scheduled maintenance and unscheduled repairs) during testing.

TOT = Total operating time (see 6.8.5).

Any support elements, i.e., technical publications, repair parts special tools, etc. , provided under the contract for assisting or conducting maintenance shall be evaluated. The system maintenance ratio shall conform to 3.14. Nonconformance with 3.14 shall constitute failure of this test.

4.6.2.10 Treatment and painting.

4.6.2, 10.0 Ferrous materiel. Application of finish 5.1.1 plus 20.24 or 24.17 of MIL-STD-171 (see 3.18). The test methods and procedures in 4.6.2.10.1.1 shall be applied to parts on the Shop Equipment, Electronic Repair, Semitrailer Mounted, composed of ferrous materiel.

4.6.2.10.1.1 Test methods and procedures.

4,6.2. 10.1.1. I Cleaning test. The cleaning test described in paragraphs under Removal of soils and corrosion product, Rinsing and Final rinse of TT-C-490 shall be performed. The total acid and alkali levels shall not exceed those specified in Rinsing in TT-C-490. This test shall be conducted daily on a minimum of four production items representative of the production process.

4.6.2, 10,1,1.2 Phosphate coating Weight test. The phosphate coating weight test shall be performed in accordance with paragraph Phosphating controls of TT-C-490. Requirements in this paragraph shall be met before the

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water break test is performed on parts. This test shall be conducted on three specimens after a maximum of four hours of phosphate processing. Test specimens shall be in accordance with Test specimens of TT-C-490.

4.6.2.10.1.1.3 water break test. The water break test shall be conducted in accordance with Condition of surface in WIL-C-53072. All requirements in this paragraph shall be met prior to painting. This test shall be conducted on all parts (100 percent) before primer is applied.

4.6.2.10.1. 1.4 Primer dry film thickness test. The primer dry film thickness test shall be conducted in accordance with paragraph Dry film thickness in MIL-C-53072. Paint thickness shall conform to the minimum values given in table V referred in this paragraph. Dry film thickness for MIL-D-23377 primer shall be .6 to .9 roils. This test shall be conducted daily on a minimum of four production items representative of the production process.

4.6.2. 10.1.1.5 Primer salt spray test. The primer salt spray test shall be performed on the primer coating. There shall be no rusting visible to the unaided eye after this test is conducted in accordance with ASTM 9 117 using a 5 percent solution for 336 hours. The items to be tested shall be representative of the items processed, i.e. , distributed randomly and processed concurrently with the production items. For a batch type paint process, four items shall be tested from each week (40 hours) of production or portion thereof. Scrap items or portions thereof may be used for this test. (Large items may be cut to a size suitable for the salt spray chamber,)

4.6.2. 10.1. 1.6 Primer and topcoat adhesion test. The primer and topcoat adhesion test shall be conducted in accordance with paragraph Adhesion of MIL-C-53072 and ASTM D 3359, method B, cross cut tape adhesion, using the 6-line pattern and 2mm spacing. Acceptance/rejection criteria shall be as indicated in paragraph Adhesion of MIL-C-53072. This test shall be perfomed on the primer coat and also after the topcoat is applied. This test shall be conducted on one item per batch, or if a continuous process is used, one item per week (40 hours, or any portion thereof'.)

4.6.2.10.1. 1.7 Topcoat dry film thickness test. The topcoat dry film thickness test shall be conducted in accordance with paragraph Dry film thickness of MIL-C-53072. Use the same test specimens that were used for the thickness test of the primer. Measurements shall be taken in the same locations as where the primer thickness was measured. The difference between the total coating thickness measurement and the primer thickness measurement is the thickness of the topcoat. Topcoat film thickness shall conform to the requirements specified in MIL-C-53072 for paint used. This test shall be conducted daily on a minimum of four production items representative of the production process.

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4.6.2.10.1.1.8 Topcoat solvent wipe test. The topcoat solvent wipe test shall be conducted in accordance with paragraph Solvent wipe of MIL-C-53072. All requirements in this paragraph shall be met prior to acceptance. This test shall be conducted on four items selected from each batch, or for a continuous process four items shall be selected from each week (40 hours of production, or and portion thereof.)

4.6.2. 10.1.1.9 Certifications and Certificates of Conformance - paint. Primer paint lots furnished with each specification shall be provided with a Certificate of Conformance verifying that the lot meets the requirements of the specification. Topcoat paint lots shall be furnished with a certification from the Belvoir Research, Development and Engineering Center, Fort Belvoir, VA, verifying that the materiel meets the near infrared reflectance properties (see paragraph Camouflage requirements and batch validation of MIL-C-53072.) Manufactured date and lot number shall be included in the information in the certification. Topcoat paint lots furnished with each specification shall be provided with a Certificate of Conformance verifying that the lot meets the requirements of the specification together with the near Infrared reflectance certification.

4.6.2. 10.2 Aluminum materiel. Application of finish 7.3.1 plus 20.24 or 24.17 of MIL-STD-171 (see 3.18). The Lest methods and procedures (see 4.6.2.10.2.1) shall be applied to parts on the Shop Equipment, Electronic Repair, Semitrailer Mounted, composed of aluminum materiel.

4.6.2. 10.2.1 Test methods and procedures.

4.6.2. 10.2.1.1 Cleaning test. The cleaning test described in paragraphs under Removal of soils and corrosion product, Rinsing, and Final rinse of TT-C-490 shall be performed. The total acid and alkali levels shall not exceed those specified in Rinsing in TT-C-490. This test. shall be conducted daily on a minimum of four production items representative of the production process.

4.6.3. 10.2.1.2 Corrosion test - pretreatment, The corrosion test-pretreatment shall be performed after pretreatment is applied in accordance with paragraph Corrosion resistance test of MIL-C-5541. Requirements in this paragraph shall be met before the water break test is performed on parts. The items to be tested shall be representative of the items processed, i.e., distributed randomly and processed concurrently with the production items. For a batch type paint process, 5 items shall be tested from each week (40 hours) of production or protions thereof. Scrap items or portions thereof may be used for this test. (Large items may be cut to a size suitable for the salt spray chamber.)

4.6.2.10. 2.1.3 Water break test. The water break test shall be conducted in accordance with Condition of surface in MIL-C-53072. All requirements in this paragraph shall be met prior to painting. This test shall be conducted on all parts (100 percent) before primer is applied.

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4.6.2.10.2.1.4 primer dry film thickness test. The primer dry film thickness test shall be conducted in accordance with paragraph Dry film thickness in MIL-C-53972. Paint thickness shall conform to the minimum lJaluP.s given in Table V referred to in this paragraph. Dry film thickness for MIL-P-23377 primer shall be .6 to .9 roils. This test shall be conducted daily on a minimum of four production items representative of the production process.

4.6.2. 10.2.1.5 Primer adhesion test. The primer adhesion test shall be conducted in accordance with paragraph Adhesion (wet) tape test in MIL-C-5541. No intercoat separation shall occur between the paint system and the conversion coating or between the conversion coating and the base metal. Any separation as stated above shall be cause for rejection. This test snail be conducted on one item per batch, or if a continuous process is used. one item per week (40 hours), or portion thereof.

4.6.2.10.2.1.6 Primer salt spray test. The primer salt spray test shall be performed on the primer coating. There shall be no rusting visible to the unaided eye after this test is conducted in accordance with ASTM B 117 using a 5 percent solution for 336 hours. The items to be tested shall be representative of the items processed, i.e.. distributed randomly and processed concurrently with the production items. for a batch type paint process, Four items shall be tested from each week (40 hours) of production or portion thereof. Scrap items or operations thereof' may be used for this test. (Large items may be. cut to a size suitable for the salt spray chamber.)

4.6.2.10.2. 1.7 Primer and topcoat adhesion test. The primer and topcoat adhesion test shall be conducted in accordance with paragraph Adhesion of MIL-C-52072 and ASTM D 3359, method B, cross cut tape adhesion, using the 6-line pattern and 2mm spacing. Acceptance/rejecticn criteria shall be as indicated in paragraph Adhesion of MIL-C-53072 This test shall he performed on the primer coat and also after the topcoat is applied. This test shall be conducted on one item per batch, or if a continuous process is used, one item per week (40 hours), or portion thereof.

4.6.2.10.2. 1.8 Topcoat dry film thickness test. The topcoat dry film thickness test shall be conducted in accordance with paragraph. Dry film thickness of MIL-C-53072. Use the same test specimens that were used for the thickness of the primer. Measurements shall be taken in the same locations as where the primer thickness was measured. The difference between the total coating thickness measurement an the primer thickness measurement is the thickness of the topcoat. Topcoat film thickness shall conform to the requirements specified in MIL-C-53072 for paint used. This test shall be conducted daily on a minimum of four production items representative of the production process.

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4.6.2.10.1.2.9 Topcoat solvent wipe test. The topcoat solvent wipe test shall be conducted in accordance with paragraph Solvent wipe of MIL-C-53072. All requirements in this paragraph shall be met prior to acceptance. This test shall be conducted on four items selected from each batch, or for a continuous process four items shall be selected from each week (40 hours) of production as portion thereof.

4.6.2.10.2.1.10 Certifications and Certificates of Conformance - paint. Primer paint lots furnished with each specification shall be provided with a Certificate of Conformance verifying that the lot meets the requirements of the specification. Topcoat paint lots shall be furnished with a certification from the Belvoir Research, Development and Engineering Center, Fort Belvoir, VA, verifying that the material meets the near' infrared reflectance properties (see paragraph Camouflage requirements and batch validation of MIL-C-53072). Manufactured date and lot number shall be included in the information in the certification. Topcoat paint lots furnished with each specification shall be provided with a Certificate of Conformance verifying that the lot meets the requirements of the specification together with the near infared reflectance certification.

4.6.2. 10.3 Ferrous and aluminum materiel Application of finish 5. 1.1 or 7.3.1 plus 20.24 of MIL-STD-171 and application of finish 5.1.1 or 7.3.1 plus 24.17 of MIL-STD-171 (see 3.18). The test methods and procedures (see 4.6.2.10.3.1) shall be applied to assemblies on the Shop Equipment, Electronic Repair, Semitrailer Mounted, consisting of mixed parts of both ferrous and aluminum materiels.

4.6.2. 10.3.1 Test methods and procedures. The ferrous parts that are installed into these assemblies shall be finished in accordance with paragraph 4.6.2. 10.1 above before assembly. The aluminum parts that are installed into these assemblies shall be finished in accordance with 4.6.7. 10.7 above before assembly.

4.7 Inspection comparison. The Government may select shops at any time during the contract production period and subject these shops to the examination specified in column 2 of table II and the tests specified in column 2 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Shops will be selected at random from those which have been accepted by the Government and shops will not include the previously inspected first zu-title model or initial production model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure o?' such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison shop to meet any requirement specified herein during and as a result of the examination and test specified (see 4.7) shall be cause for rejection of the inspection comparison shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on shops previously produced and accepted under the contract. Any deficiencies found as a result of inspection comparison inspection will be considered prima facie evidence that all shops accepted after approval of the initial production shop are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Packaging inspection. Packaging inspection shall be conducted to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Level C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Preservatives. Preservatives specified herein shall conform to MIL-P-116.

5.1.1.2 Methods. Methods of preservation and packaging specified herein shall conform to MIL-P-116.

5.1.1.3 Sets or kits. When items are furnished in sets or kits with a carrying case, the items shall be preserved as specified herein and, when possible, replaced in the case.

5.1.1.4 Items having mounting brackets or holders. Items furnished with mounting brackets or holders shall be preserved as specified herein and, when possible, replaced in the bracket or holder and secured to prevent movement and damage.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces of the components, not specifically specified herein, requiring the application or a contact preservative in accordance with MIL-F-116 shall be preserved as follows:

5.1. 1.5.1 Unfinished surfaces. Unfinished surfaces shall be coated with P-1 preservative.

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5.1.1.5.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with P-6 or F-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type 1, grade A, class 2 and secured in place with tape conforming to PPP-T-60, type IV.

5.1.1.6 Semitrailer. The semitrailer and mounted van body shall be preserved in accordance with the Level A (mobile) preservation and packaging requirements of MIL-V-62038.

5.1.1.7 space heater. The heater shall be completely drained of fuel. Exterior body openings shall be sealed with tape conforming to PPP-T-60, type TV, or with barrier material conforming to PPP-B-1055, class E-2 H-1 or H-5, secured in place with tape of PPP-T-60, type IV.

5.1.1.8 Air conditioner. Exterior body openings shall be sealed as specified in 5.1.1.7 for the space heater.

5.1.1.9 Handtools and tool sets. Handtools and tool sets shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.10 Soldering equipment. Soldering equipment shall be preserved in accordance with the Level A requirements of MIL-W-45562 except unit containers shall conform to nonweather-resistant class.

5.1.1.11 Hardware. Hardware such as bolts, nuts, screw, and washers shall be preserved in accordance with the Level A requirements of PPP-H-1581.

5.1.1.12 Items furnished in commercial containers. Items and materials such as lubricating oil, silicone compound, grease, etc., commercially furnished in sealed can, pails, drums, jars, or tubes shall not require additional preservation or packaging.

5.1.1.13 Method III. Items such as antenna elements, battery filler brackets, stools, small part cabinets, and ground rods shall be preserved Method III.

5.1.1.14 Method IC-1 or IC-3. Items such as electrical clips, electrical plugs, electrical clamps, electrical sleeving, jacks, extension lights, mine detecting test pins, and probes shall be preserved Method IC-1 or IC-3. Items of like description shall be packaged together when practical.

5.1.1.15 Method IC-2. Items such as the first aid kit, insulation tape, battery tester and leakage testers shall be preserved Method IC-2. The container shall be the contractor's commercial container or a close-fitting box conforming to PPP-B-566 or PPP-B-636, class domestic.

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5.1.1.16 Method IIe. Items such as battery chargers, vacuum cleaner, multimeters, oscilloscopes, oscillators, voltmeters, wave meter, bridge converter, signal generator frequency measuring set, power supply, test set, and power meters shall be preserved Method Tie. Each item shall be contained within the carrying case, when one is furnished, the contractor's commercial container, or in a close-fitting box conforming to PPP-B-536, class domestic.

5.1.1.17 Electrical wire. Electrical wire shall be packaged in accordance with the Level A preservation and packaging requirements of MIL-C-12000.

5.1.1.18 Cable assemblies and test leads. Open ends shall be covered or sealed to provide protection against foreign material. Lengths not on reels shall be coiled to a safe minimum diameter. The coils shall be uniform, compact and of a diameter to prevent deformation or kinking. Each coiled cable shall be secured a minimum of three equidistant places with cotten twine or pressure sensitive tape. Loose fittings shall be placed in a transparent plastic bag, sealed and secured on the inside of the coil.

5.1.1.19 Generator set (MAP). When furnished, the generator set shall be preserved, packaged, packed, and marked in accordance with the Level A requirements of MIL-G-28554.

5.1.1.20 Locks and keys. The interior of locks shall be coated with dry lubricant conforming to MIL-M-7866. The locks shall be operated to assure penetration of the lubricant. The locks used for securing the doors shall be protected against the elements by wrapping the locks in barrier material conforming to MIL-B-121, grade A. The wrap shall be secured with tape conforming to PPP-T-60, type IV. The keys shall be preserved with method III and forwarded to destination separately by registered mail. A cloth or paper shipping tag containing the shop equipment identification number and the trailer registration number shall be attached to the related padlock keys for future positive identification.

5.1.1.21 Fire extinguisher. Unpainted exterior metal surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-p-116 shall be coated with P-1 preservative. The extinguisher shall be placed in the mounting bracket and secured.

5.1.1.22 Electronic and electrical equipment. Electronic and electrical equipment and accessories, shall be preserved in accordance with MIl-E-17555, level A and, when applicable, as specified in 5.1.1 .16.

5.1.1.23 Grinder. The grinder shall be preserved in accordance with MIL-M-18058, Level A.

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5.1.1.24 Stools. The stools, disassembled to the maximum practical extent and preserved as specified in 5.1.1.5, shall be packaged in a fiberboard box conforming to PPP-B-636, class domestic. Cushioning shall be in accordance with MIL-STD-1186,

5.1.1.25 Miscellaneous supplies and equipment. Items, equipment, and components not specifically identified herein shall be preserved as specified for supplies and equipment of similar design, construction, and characteristics.

5.1.1.26 Consolidated packaging. Components that cannot be placed in drawers, toolboxes, compartments, or secured by the mounting brackets or holders provided shall be consolidated together in close-fitting boxes conforming to PPP-B-601, domestic type. The boxes shall be of a size to fit into available space. The contents shall be blocked, braced, and cushioned as applicable to prevent movement and damage to the contents within the box.

5.1.2 Level C. Except as specified in 5. 1.2. 1 , Level C preservation and packaging shall be in accordance with MIL-STD-1190 as specified herein. Components shall be placed and secured in the compartments provided. Toolboxes, drawers, mounting brackets, and holders shall be used to the maximum extent.

5,1.2,1 Gas engine. The engine shall be preserved in accordance with Level A requirements of MIL-E-10062, using the alternate procedures for preserving the combustion chambers and valves.

5.2 Packing See 6.9). The semitrailer shop, with equipment enclosed, shall be prepared for mobile shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings. Items and equipment installed in the shop body, and other movable equipment or items, shall be cushioned, blocked, braced, and anchored to prevent damage to the equipment or to the body. All doors and drawers shall be secured to prevent accidental opening anti to prevent pilferage of contents. When there is not suficient space within the body, the generator set shall be shipped separately from, but at the same time as, the semitrailer shop.

5.3 Marking. All marking requirements shall be in accordance with MIL-ST'D-129. Barcode marking is required and shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended usc. The electronic repair shop is intended to be used as a mobile shop with a set of hand and machine tools, electronic test instruments, and work space for electronic repairmen repairing military electronic equipment in the field.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).
(c) First article when required (see 3.1).
(d) When initial production inspection is required and number of shops to be furnished (see 3.3).
(e) Government-Furnished property if different (see 3.19). (f) Government-loaned property if different (see 3.20).
(g) Preservation as required (see 5.1).

6.3 Set components list. The contracting officer should arrange to furnish information concerning items specified in drawing A7551983 (see 3.2).

6.4 Technical interface data. The contracting officer should arrange to provide information concerning technical interface for the Government furnished semitrailer specified in 3.19

6.5 Lubricants. When requested by the contractor, the contracting officer should obtain a list of military lubricants applicable to the shop covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.15).

6.6 Government-loaned property. The contracting officer should arrange to loan the contractor the property specified in 3.20.

6.7 Government-furnished property. The contracting officer should arrange to furnish the supplier the property specified in 3.19.

6.8 Definitions.

6.8.1 Maintenance ratio. The ratio of the number of maintenance man-hours required for both scheduled maintenance and unscheduled repairs to the number of hours of operation (see . 2. 14).

6.8.2 Displacement. Displacement is defined as dislodgement of any tool or item of equipment from its designated position. Movement of a tool or item of equipment within its designated position without any dislodgement is permitted (see 3.10.2 and 3.14).

6.8.3 Chargeable reliability failure. A chargeable reliability failure is defined as any malfunction:

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a. Which cannot be corrected by organizational repair or replacement action using the controls on equipment tools or parts.
b. Which causes or may cause:

(1) Failure to start operation.
(2) Cessation of operation
(3) Degradation of performance below specified levels.
(4) Damage to the shop by continued operation.
(5) Safety hazards to personnel.

6.5.4 Mean-time-between-failure (MTBF)- For a particular interval, the total functioning life of a population of an item divided by the total number of failure within the population during the measurement interval. The definition holds for time, cycle, miles, events, or other measure or life units (see 4.6.2.7).

6.8.5 Operting time (OT). Defined to be an 3-hour period per mission. This time does not include scheduled maintenance or unscheduled repair time If the mission profile is completed in less than 8 hours, the balance is considered to be standby time (see 4.6.2.7 and 4.6.2.9).

6.8.6 Corrective maintenance action. Any test incident that requires unscheduled maintenance time and includes both reliability chargeable failures and nonchargeable incidents (see 4.6.2.8).

6.8.7 Mean time to repair (MTTR). The total corrective maintenance time in clock hours divided by the total number of corrective maintenance actions (see 4,6.2.8).

6.8.8 Slave starting. External electrical power applied to a generator whose batteries have become discharged/depleted which may receive electrical battery voltage from an outside source through a slave cable plugged into the slave receptacle.

6.9 Packing (see 5.2). Levels of packing have not been included for semitrailer and mounted equipment as the nature of the item is such that only mobile packing is appropriate. For purposes of preservation and packing markings, the level of packing should be considered level A and should be marked A/A.

6.10 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporaed in the contract.

6.11 Provisions. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of shops.

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6.12 Ear protection. When the noise level exceeds category D of MIL-STD-1474, the operator's manual for the shop shall include appropriate instructions for use of hearing protection when using equipment that produces noise above 85 decibels (dB) (A) inside and outside the shop, and within a distance from the shop at which 85 dB(A) will be met.

6.I3 Subject term (key word) listing.

Environmental
Readability
Tools and equipment
Maintainability
Mission
Reliability

6.14 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian:
Army - AL

Preparing activity:
Army - AL

Project No. 4940-A612

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-52377D(AL), Shop Equipment, Electronic Repair, Semitrailer Mounted

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

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